STHIROS
Strategy. Risk. Cyber. AI.



Taking a manufacturer's intelligence from a 35 to one governed Smart Genba view

A leading automotive component manufacturer engaged Sthiros to architect a Smart Genba data foundation, unifying 25 datasets and cutting BI dead inventory by 90.4% across its ecosystem.

Client	Service	Sector	Geography
A Leading Automotive Component Manufacturer	**Manufacturing Intelligence Data Architecture & Governance**	**Automotive, Manufacturing**	**India**

The Challenge:

When 25 datasets can't agree on one truth

This manufacturer ran a mature Microsoft estate, Oracle ERP, Azure, Power BI, spanning IT and OT systems, but scored just 35 on its Manufacturing Intelligence Quotient. Data existed everywhere, yet no governed semantic layer connected production, quality, and maintenance.

Fragmented Data Islands
IT and OT systems generated data everywhere, with no unified connections.

Failing Data Pipelines
View-on-view dependency chains forced re-traversal of 24 months of data every query.

Ungoverned BI Delivery
Publish to Web exposed reports without authentication, a live security breach point.

No Semantic Layer
Decentralised datasets meant zero asset reuse and conflicting metric definitions.

Growth Constraint
Without a governed foundation, Jidoka and predictive maintenance stayed out of reach.

BI Dead Inventory
Report sprawl buried the few assets that actually drove business decisions.

The Solution:

Five pillars, replacing fragmented data with one governed model

Sthiros architected a governed data foundation across five pillars, from a single semantic model to real-time alerting, rolled out in a phased Crawl, Walk, Run sequence rather than all at once.

Every KPI, OEE, MTBF, and Rejection PPM among them, now runs off one governed model instead of department-by-department logic.

The Methodology

1

Semantic Layer Blueprint

A governed enterprise semantic model codified core dimensions and fact tables, ensuring a single version of the truth for OEE, MTBF, and Rejection PPM.

2

Metadata-Driven Pipelines

Manual, custom-built integrations replaced with a configurable orchestration framework, ingesting eight distinct IT and OT categories through one configuration table.

3

ADLS Gen2 Delta Lake

Mixed Blob containers decommissioned in favour of a governed Medallion structure, using CETAS materialisation and Spark Delta conversions to break dependency chains.

4

Security Governance

Publish to Web removed in favour of Power BI Embedded delivery, with role-based and row-level security ensuring only authenticated access to sensitive data.

5

Alert-to-Action Framework

Power Automate integrated with the semantic layer to trigger real-time notifications the moment a KPI threshold is breached.

What This Means

One number for every team.

Cut storage costs by 90%.

Clear the clutter before AI.

Outcomes

50-90%

Storage costs reduced

Automated lifecycle tiering moved historical data from Hot to Archive tiers.

90.4%

BI dead inventory cut

302 unused reports archived, focusing compute on the 9.6% that drive value.

25

Datasets unified

One single semantic model now governs Production, Quality, and Maintenance.

161M+

Rows eliminated from re-extraction

Incremental Delta Lake materialisation ended daily full re-extraction of Oracle ERP data.

Get in touch

Is your shopfloor data trusted, or just generated?

Speak to Sthiros about what a governed data architecture can unlock for your Smart Genba journey.

Connect with us:
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